

A Guide to Testing Web Applications

DevFest 14th Dec 2019 Bishkek

- Alan Richardson
- [eviltester.com/conference/
devfestbishkek2019_conference](https://eviltester.com/conference/devfestbishkek2019_conference)
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Have you ever wondered how other people test applications? Not in theory, but in practice? What thought processes are used? How did they model the application? What tools were used? How did they track the testing? That's what this talk is all about. This talk will be based on a short Case Study of testing an open source web application. Why open source? Because then there is no commercial confidentiality about the process, tools or thought processes.

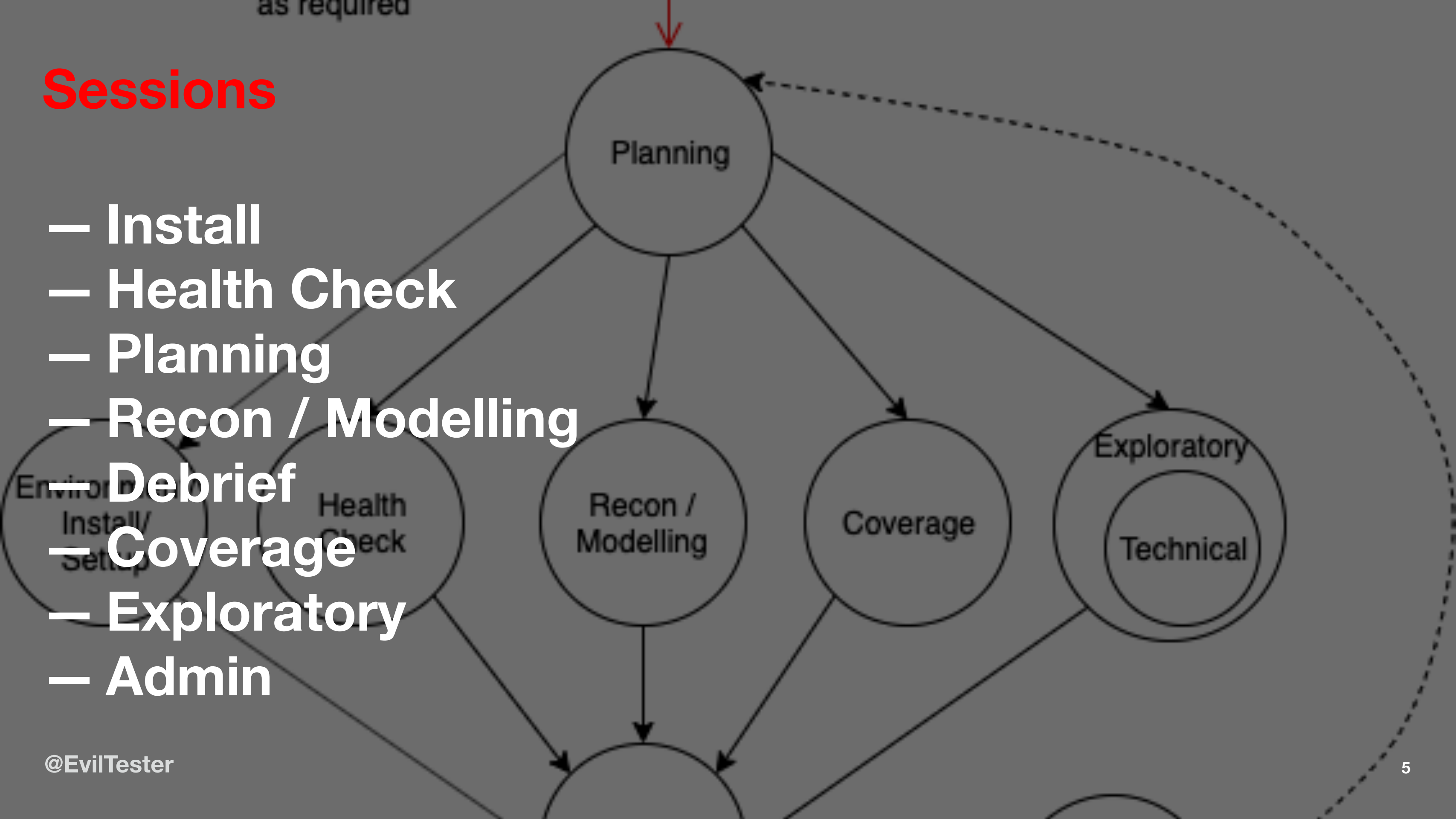
Alan will explain his thought processes, coverage, approaches, tools used, risks identified and results found. And generalise from this into reusable models and principles that can be applied to your testing. This covers the What?, and the Why? of practical exploratory web testing.

A 'micro' case study

- Short Case Study
 - about a 'day' of Testing
- Lessons and observations are extrapolated to a 'macro' level
- not all Testing Approaches
- lessons mentioned, have context

Sessions

- Install
- Health Check
- Planning
- Recon / Modelling
- Debrief
- Coverage
- Exploratory
- Admin



Which App to choose

- Tracks
- <https://www.getontracks.org/>
- Web App, JS, API, DB
- Offers a lot of scope for testing

contexta

☆ □ ✎ ▼ contexta todo

contextb

☆ □ ✎ ▼ uncomplete

contextc

☆ □ ✎ ▼ contextc todo

con

☆ □ ✎ ▼ Action

Completed actions in this context

☆ ☒ ▼ 27/11/2019 contextb todo (contextb)

Show all

Send feedback on 2.3.0: [Bugs](#) | [Wiki](#) | [Mailing List](#) | [Website](#) | [Contribute](#) | [Mobile Site](#)
Tracks Appliance – Powered by [TurnKey Linux](#)

Hide form | Add multiple next actions

Description



Notes

Project

Context

contexta

Tags (separate with commas)

Due

Show from

Depends on

✓ Add action

Active projects (0)

- None

Active contexts (4)

- contexta (1 action)

Lesson

- Normally we don't get to choose
- Allocate staff based on skills and experience,
- If we don't understand the technology
 - we need to manage that as a risk

Environment

Could use:

- Docker
- Local install
- VM
- Cloud hosted

Different risks and constraints.

My Environment Requirements

- **proxy traffic**
- **possibly view DB**
- **low impact on my machine**

Knowledge Constraints

- My Tech knowledge of Docker
- Lower Docker knowledge than VM
- Know how to configure and access DB on VM
- Chosen VM <https://www.turnkeylinux.org/tracks>
- Issue... old version of tracks
 - not a concern for this talk

Lesson Environment Impacts Testing

- Do staff have permissions?
 - to observe, interrogate, manipulate?
- Does environment match deployment?
- Does release process match live?
- Version management
 - same as live?
 - can do upgrades?

Install Session

Install Session

20191125 10:00

* download vm

<https://www.turnkeylinux.org/tracks>

* install tracks

* 192.168.1.36

* R00troot

10:30

Environment Ready

- **Installed VM**
- **Login as Admin**
- **Is tracks ready?**
 - **Don't know**
 - **No health check page**
 - **no status page**

Lesson

- Automated Execution helps derisk deployment
- Otherwise first 'test' sessions are 'health check'
- Risk of wasted time but finding out too late

Health Check

- **CRUD**
- **Create, Read, Update, Delete**
- **Main Areas**
 - **Keep it simple, create a todo**

Health check Session

20191125 - 14:00

- Create from front page

- [Add Action] - blank, XHR, error message rendered via JavaScript interpreted from Response
- [Add Action] - Desc 'a' - error, requires Context
- [Add Action] - Desc 'a', Context 'c' - created

- Read by changing between tabs

- home -> projects -> home and todo rendered

- Amend from front page

- a -> aa

Delete from front page

- create 'b' in 'c' context, for deletion
 - marked as completed anticipating delete option but moved to completed
 - Note - completed action can be dragged ?To where, for what purpose?

contexta

    contexta todo

contextb

    uncomplete

contextc

    contextc todo

con

    Action

Completed actions in this context

   27/11/2019 contextb todo (contextb)

Lessons

- All testing is exploratory, observation
- Exploration boosts (and is constrained by) technical knowledge of the app
- Time Box Testing
- Need screenshot tools
 - gather evidence
- Notes Prior (Plan) (AIMs)
- More Notes During (Actual)

How do I know what to test?

- Requirements
- Stories
- Defects
- Change Requests
- Commits

Planning Session

- Making the decision about what to test
- Planning: exploratory process of weighing up factors, asking questions and making decisions
- Tacked and planned like any other activity
- do not go too deep
 - working with unknowns
- No plan - lose an opportunity
 - to effectively target our testing.

Planning session

20191126 - 13:00

- release notes
- commits?
- defects?
- general functionality?
- assume team tested, add value by going 'holistic'
- pick release note item and go beyond acceptance criteria

"You can now change the state of a context to closed"

Questions:

- What is a Context?
- What is Context State?
- What states are there?
- Is it a state machine?
- Are all states equally valid?
- Are there transition rules?

20191126 - 13:10

My Chosen Scope

From <https://www.getontracks.org/news/comments/release-2.3.0/>

"You can now change the state of a context to closed"

menu). This also works for tag page, the project page, the tickler and the context page

- You can now change the state of a context to closed
- Czech locale has been renamed from cz to cs to follow ISO standards
- Added Russian locale (@AlexStein)
- The toggle-notes and toggle-collapsed-containers have been moved into the view menu.

I am going to

- Test Holistically**
 - System rather than Story**
- Assume other testing has been performed**

When I start testing, I don't know what that means yet.

I have to explore to find out.

Recon Session

Recon Session

Aim - Understand and model

> "You can now change the state of a context to closed"

- What is a Context?
- What is Context State?
- What states are there?
- Is it a state machine?
- Are all states equally valid?
- Are there transition rules?

20191127 12:22

192.168.1.36

Recon Session Findings

- Actions added to context
- context seems to move between state in any order (TODO: cover state table)
- context constraints
 - Not close if open actions
 - not add open actions to closed
- But, can amend actions
 - to be on closed contexts

Recon Session Findings

- all contexts are shown in auto complete drop down regardless of state (order controlled by Organize view)
- seems to be XHR updates
 - but need to view traffic to be sure

Lessons Learned

- **Functional Testing is Functional Testing**
- **Architecture & technology -> observation ability**
- **My modelling and assessment of findings is limited by my ability to observe**

Tooling Lessons Learned

- **Selecting Tooling is an evolving process based on the needs of the testing and the application.**
- **I don't 'start' with tools, I iterate towards tool usage as required.**

Debrief Session

- Look at what we've done
 - Working exploratory means I have to 'step back' periodically
- Collate Notes
- Identify Issues, Todos, Defects
- Feed into Planning Session
- Refine/Formalise Models
 - Modelling Session if 'large'

Debrief Session

20191128 - 10:40

Aim: review notes, identify todos, issues, models, status, summarise

- Have installed software in VM, seems to work well enough for testing
- Gui rendering issues on pop up drop downs
 - in situ on screen
 - when dragging on screen
 - need to have screen capture tool ready to capture rendering issues as they vary between sessions
- Learned basic mandatory relationship between Action and Context
 - Action needs a Context
- Currently observing results on GUI, want to expand observation to traffic and DataBase.

Lessons Learned

- I rely on linear text as much as possible
- text files, markdown, screenshots in `/images`
- formatting, aesthetics, structure - later
- essential when working on own
- can act as a 'status' report

Modelling Session

- "prep a more comprehensive coverage approach because I didn't 'test' this I did an initial exploration during recon but the coverage wasn't documented or thought through well enough."

Lessons Learned

- **Modelling is a key skill**
- **If models are ambiguous, we don't understand**
- **Simple concepts expand into complex models**
- **Complex models -> high number of combinations**

Lessons Learned

- Informal modelling can support exploration
- Don't leap to a diagrammer tool
 - pen, paper, smartphone camera
 - whiteboard -> smartphone camera
 - tech solution smartpen -> Evernote

Led to a Quick Planning Session

Most important to explore

CONTEXT - CLOSED

CONTEXT OPEN $\rightarrow$ CLOSED

CONTEXT CLOSED $\rightarrow$ OPEN

ACTION $\leq$ ACTION
EVENT STATE

- CREATED $\leftarrow$ NORMAL
- CHANGED CONTEXT
- CLOSED
- OPEN
- DEPENDENCIES

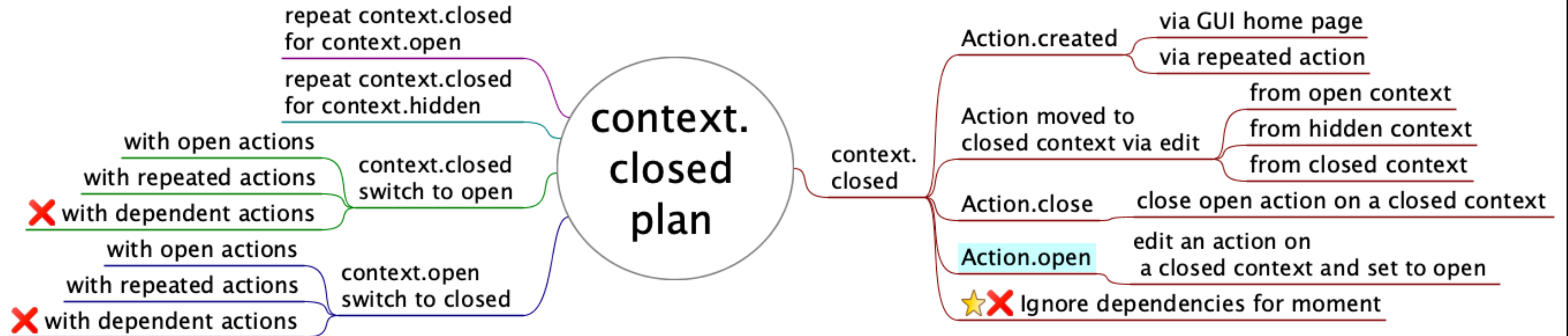
Lessons Learned

- **Modelling and Planning are related**
- **Planning is prioritising parts of model coverage**
- **Plan only as far as you need to to support the next session**

Coverage Session

- Actual Testing
- Mindmap in advance to expand the plan
- kept notes as tested
- missing items from the 'plan'
 - added them into my notes
- minimise additional exploration
 - focus on 'coverage'
 - follow later

Expanded Plan



Coverage Session Notes

Coverage Session

20191128 13:10

13:16 - expanded plan saved to ./context-closed-expanded.plan.mm

edit <http://192.168.1.36/contexts/1> to be named context.closed

Initially 0 Actions

5 Active contexts

Context name
context.closed

Context state
☐ Active ☐ Hidden ☒ Closed

☒ Update ☐ Cancel

Lessons Learned

- Gathered more evidence, than during recon
- Expand plans as required
 - FreePlane because of build in scripting
- Plan will not be complete
 - awareness of new opportunities
 - follow (inscope), or defer

(Technical) Exploratory Session

"Aim: Investigate the traffic mechanisms for creating actions and changing state on context - how far can we push this?"

Technical Exploratory Session Difference From Exploratory Session

- Observation at a technology level
 - e.g. traffic, params, headers
- Manipulation at a technical level
 - e.g. resend requests, DOM manipulation, bypass GUI, Tools mandatory
- Technical observation -> new ideas

Using

- Firefox (fewer plugins to create noise)
- FoxyProxy plugin (configure Proxy direct)
- Firefox Dev Tools (interrogate/manipulate DOM)
- Owasp ZAP Proxy (Observe, Interrogate, Manipulate HTTP)
- TextEditor - view XML
- SnagIt - screenshots
- Evernote - note taking

- ### Active contexts
- contexta (1 action)
 - contextb (1 action)
 - contextc (1 action)
- ### Hidden contexts
- context.closed.rename (5 actions)
 - contextd (1 action)
- ### Closed contexts
- new closed.context (2 actions)
- Send feedback on 2.3.0: [Bugs](#) | [Wiki](#) | [Mailing List](#) | [Website](#) | [Contribute](#) | [Mobile](#)
- Tracks Appliance - Powered by TurnKey Linux

Inspector Console Debugger Style Editor

Search HTML

```
<a id="link_edit_context_1" class="context_edit_settings icon" href="/contexts/1/edit">...</a>

<div class="context_description">
  <a href="/contexts/1" title="View context: context.closed.rename">context.closed.rename</a>
  (5 actions)
</div>
</div>
```

Rules Layout Computed Ch

Filter Styles :hov .cls +

```
text-decoration: none;
padding-left: 1px;
padding-right: 1px;
Inherited from body
...c2afd7f76c40d1d7f6735e6a3cc6d.css:5
body {
  font-family: "Lucida Grande", Verdana, Geneva, Arial, sans-serif;
  font-size: 80%;
```

Contexts-hidden.ui-sortable > div#container_context_1.list > div#context_1.cor

tracks_data.xml

```
<created-at type="dateTime">2019-11-27T12:36:02+00:00</created-at>
<updated-at type="dateTime">2019-11-27T15:56:03+00:00</updated-at>
<state>active</state>
</context>
<context>
  <id type="integer">6</id>
  <name>new closed.context</name>
  <position type="integer">6</position>
  <user-id type="integer">1</user-id>
  <created-at type="dateTime">2019-11-28T13:31:04+00:00</created-at>
  <updated-at type="dateTime">2019-11-29T12:21:56+00:00</updated-at>
  <state>closed</state>
</context>
</contexts>
<nil-classes type="array"/>
<nil-classes type="array"/>
<nil-classes type="array"/>
<nil-classes type="array"/>
<recurring-todos type="array">
  <recurring-todo>
    <id type="integer">1</id>
    <user-id type="integer">1</user-id>
    <context-id type="integer">1</context-id>
    <project-id type="integer" nil="true"/>
    <description> repeated action in closed context</description>
```

Standard Mode

Untitled Session - 20191129-1211 - OWASP ZAP 2.8.0

Sites +

- https://192.168.1.36
- assets
- contexts
- 1
- GET:1
- POST:1(_group_view_by,_method,_source_view,...)
- 2
- GET:6
- GET:contexts
- GET:contexts.autocomplete([object Object],term)
- GET:contexts.autocomplete(term)
- data
- GET:data
- GET:login
- POST:login(authenticity_token,login,user_login,user_...
- GET:preferences
- todos

Quick Start Request Response

Header: Text Body: Text

POST https://192.168.1.36/contexts/1 HTTP/1.1

User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS 10_15_0) Gecko/20100101 Firefox/70.0

Accept: text/javascript

Accept-Language: en-GB,en;q=0.5

X-CSRF-Token: 6zkIFBkNF9u6Xax5qtJGE06YQfzzCV6NSp6%2B6l8%2Fm5c...

Content-Type: application/x-www-form-urlencoded

X-Requested-With: XMLHttpRequest

Content-Length: 245

Origin: https://192.168.1.36

_group_view_by=context&_method=put&_source_view=...&authenticity_token=6zkIFBkNF9u6Xax5qtJGE06YQfzzCV6NSp6%2B6l8%2Fm5c...&context%5Bname%5D=context.closed.rename&context%5Bstate%5D=hid...

History Search Alerts Output WebSockets

Timestamp	Method	URL	Code	Reason	RTT	Size	Resp. Body	Highest Alert	No
29/11/19 12:33:45	GET	https://192.168.1.36/contexts	200	OK	32...	16,010	bytes	Low	
29/11/19 12:33:49	GET	https://192.168.1.36/contexts/1/edit?_sour...	200	OK	23...	2,240	bytes	Low	
29/11/19 12:37:07	POST	https://192.168.1.36/contexts/1	200	OK	31...	2,192	bytes	Low	
29/11/19 12:37:59	GET	https://192.168.1.36/contexts/1/edit?_sour...	200	OK	32...	2,243	bytes	Low	
29/11/19 12:38:18	POST	https://192.168.1.36/contexts/1	200	OK	32...	2,187	bytes	Low	
29/11/19 12:39:50	POST	https://192.168.1.36/contexts/1	200	OK	58...	2,264	bytes	Low	
29/11/19 12:39:59	GET	https://192.168.1.36/contexts	200	OK	37...	16,034	bytes	Low	
29/11/19 12:40:29	GET	https://192.168.1.36/contexts/1/edit?_sour...	200	OK	37...	2,248	bytes	Low	
29/11/19 12:41:48	POST	https://192.168.1.36/contexts/1	200	OK	32...	424	bytes	Low	
29/11/19 12:42:01	GET	https://192.168.1.36/contexts	200	OK	45...	16,031	bytes	Low	
29/11/19 12:42:04	GET	https://192.168.1.36/contexts/1/edit?_sour...	200	OK	27...	2,247	bytes	Low	
29/11/19 12:43:11	POST	https://192.168.1.36/contexts/1	200	OK	26...	424	bytes	Low	
29/11/19 12:44:11	POST	https://192.168.1.36/contexts/1	200	OK	29...	424	bytes	Low	
29/11/19 12:45:41	POST	https://192.168.1.36/contexts/1	200	OK	56...	2,208	bytes	Low	
29/11/19 12:46:39	GET	https://192.168.1.36/contexts	200	OK	48...	16,031	bytes	Low	
29/11/19 12:47:32	POST	https://192.168.1.36/contexts/1	200	OK	33...	2,261	bytes	Low	
29/11/19 12:47:36	GET	https://192.168.1.36/contexts	200	OK	52...	16,031	bytes	Low	
29/11/19 12:48:40	POST	https://192.168.1.36/contexts/1	200	OK	26...	424	bytes	Low	
29/11/19 12:50:33	POST	https://192.168.1.36/contexts/1	200	OK	27...	2,221	bytes	Low	

Alerts 0 0 5 2

Current Scans 0 0 0 0

Basic Process

- Work in GUI. Observe impact via HTTP (I can see if JS or Server validation used)
- Interrogate HTTP Traffic, source of new ideas
- Manipulate (resend) HTTP Traffic
- Repeat HTTP results via DOM manipulation
- keep an eye on time and scope

Notes Overview

20191129 11:50

Exploratory Session

"Aim: Investigate the traffic mechanisms for creating actions and changing state on context - how far can we push this?"

- Install new version of ZAP Proxy - v2.8.0
- refreshed certificate in firefox - preferences, certificates, "trusted issuer" - choose 'options' in import to select
 - had to delete existing, before it all worked
 - ZAP hud started when I went to site. I just want basic proxy at the moment so configured to switch it off

12:05 - proxy ready

restarting session

On zap on Mac - tried to persist session in sub folder but it doesn't show folders - is that a bug or an odd mac thing after OS upgrade - not sure if session is actually being persisted - saving to ~/Documents/currentdate

Notes Subset

- create a new Action on a closed context

Using 'new closed.context'

"create 1128 Action on closed context"

url encoded form post

POST <https://192.168.1.36/todos>

utf8=%E2%9C%93&authenticity_token=6zkIFBkNF9u6Xax5qtJGE06YQfzzCV6NSp6%2B6l8%2Fm5c%3D&default_project_name=&default_context_name=contexta&new_todo_starred=false&todo%5Bdescription%5D=create+1128+Action+on+on+closed+context&todo%5Bnotes%5D=&project_name=&context_name=new+closed.context&initial_tag_list=&tag_list=&todo%5Bdue%5D=&todo%5Bshow_from%5D=&predecessor_input=&predecessor_list=&_source_view=todo&_source_view=todo&_group_view_by=context&_tag_name=

what is default context name used for?

Looked in preferences and can see a default context for email - set to context.closed (not sure how that happened)

Can't see a setting for default context.

What if I pass in the message without a context_name but have the closed context as the default_context_name?

What I Found

- **ZAPProxy** - session only saveable to Documents, not sub folder
- **Tracks no error message when incorrect status used for context, value ignored, 200 OK**
 - suggests Backend 'trusts' the GUI
 - investigate error handling

What I Found

- Tracks XHR message responses which are error messages are 200 status rather than a 4xx status
- 'interesting things about tracks' investigate later
 - `_method=put` field differs HTTP verb,
 - HTML and JS response to XHR
 - "default context" field

What I Found

- **API vs GUI mismatch? a forms API and a JSON API**
- **limited observation - did not include DB tools, had to use System exports to check data.**

Tooling Benefits

- dev tools for DOM Inspection and manipulation
- automatically record HTTP session
- proxy HTTP Observation, Inspection & Manipulation
- easy replay for requests
- easier traffic inspection

Lessons Learned

- Tools can impact testing - lost time tool setup
- More information -> more risk of going off charter or off time
- Test Ideas that are not possible without tooling
 - e.g. duplicate params,
 - re-order params

Lessons Learned

- Automated record keeping is a 'backup', I still copy HTTP request and URLs into my notes

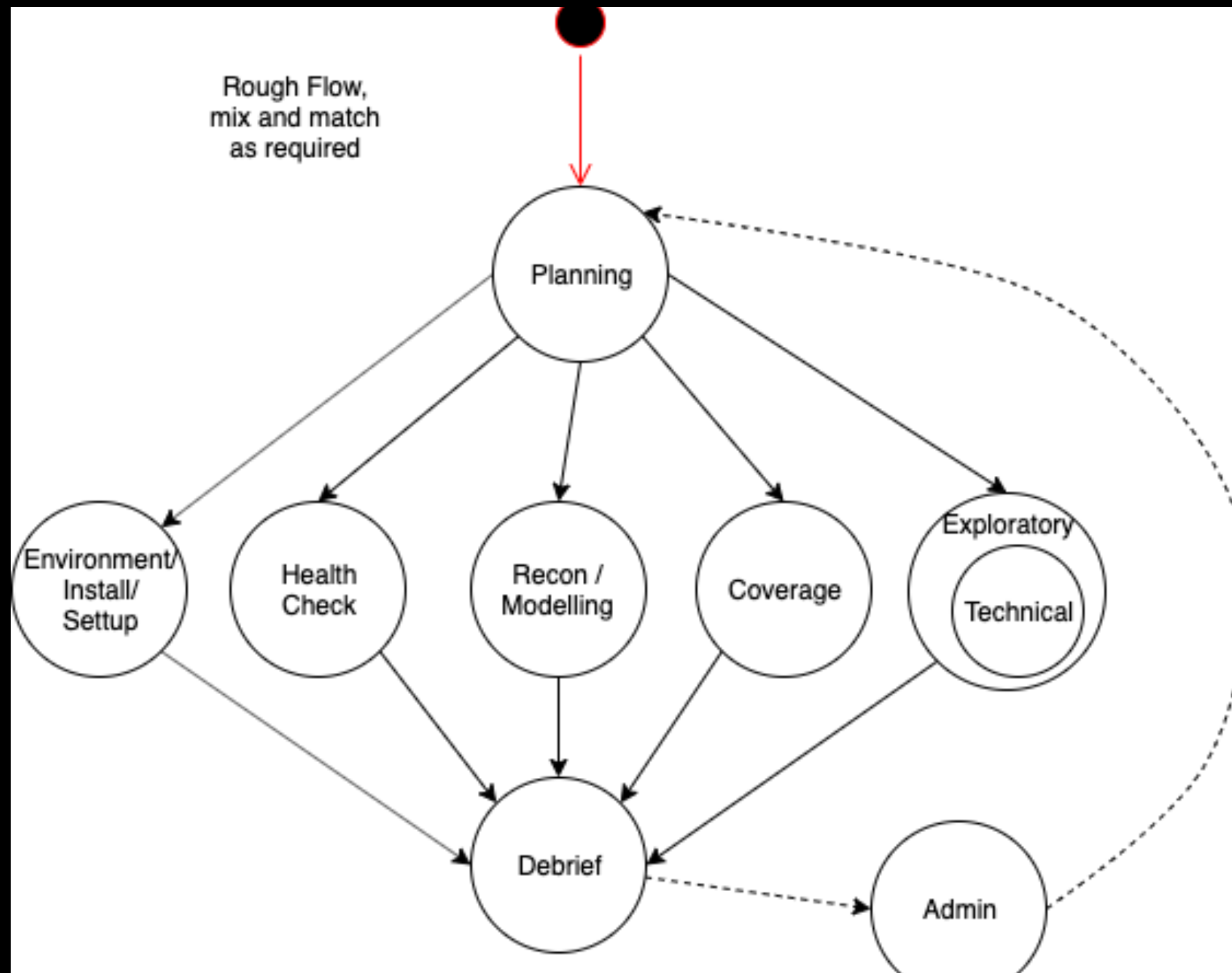
What Did I Not Do/Show? - Automating

- likely factored into "Story" Testing
- start tactical, make it work, refactor to abstraction layers, become more strategic as required
- use default tools
 - get value out of quickly
 - new tools as required

What would I do next (for current scope)?

- **API**
 - repeat GUI and HTTP at API
 - assumption - different routing code
 - possibly different back end code
- **DataBase**
 - observation and interrogation

Basic Testing Flow Used



Testing

- 241 minutes (4 hours) spent Testing
- 93 minutes (1.5 hours) hands on
- 2-3+ hours additional admin if 'real' project

Much of Testing goes unnoticed and untracked.

Note taking

- key to gathering evidence
- and making testing visible.

Tooling comes from:

- need
 - observation
 - interrogation
 - manipulation
 - admin, evidence
- technology
 - can aid/hinder tooling

We explore at all points in our testing

- The more we are focussed on coverage**
 - the more we constrain our exploration to the Coverage.**
- The more we are focussed on Exploration,**
 - the more our coverage has to be reverse engineered from:**
 - our logs,**
 - evidence, and notes.**

About Alan Richardson

- EvilTester.com
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Social & Contact Links

1. <https://www.eviltester.com>
2. <https://twitter.com/eviltester>
3. <https://www.linkedin.com/in/eviltester/>
4. <https://www.youtube.com/user/EviltesterVideos>
5. <https://instagram.com/eviltester>
6. <https://facebook.com/eviltester>

Section: Edited for length

The slides that follow were not used in the main presentation due to timing constraints.

Recon Session Findings

- open screens to not update with state unless refreshed e.g. home, when open and move context to new state, does not show new state - can this be used to abuse state? e.g. drag todo as sub todo on a todo on a closed state and closed todo that has not refreshed yet?

If this was a real project: Planning

- add this as notes in a Jira task**
- add this to kanban board**
- discuss with team**
- possibly expand further and write up, if not doing immediately and model is complex**

Tooling Lessons Learned

- tooling support for observation, interrogation.
manipulation appropriate for scope and aim
- i.e. I didn't use any extra tools for testing, but I did
for planning

In-Sprint, Story and System Testing

In-sprint testing

- constrained to 'work done' scope
- artificially constrained - no error handling, no security etc.
- need to track these 'omissions' as coverage gaps for later testing
- often testing 'bits' of stories

Stories

- Complete chunk of work
- Sometimes span sprints
 - 'not complete' but still requiring testing
- Acceptance Criteria testing needs to be revisited
- Automated Execution to cover criteria can help
- Going 'beyond' the story often needs to wait till story is 'complete'
- Stories often tested in isolation - Automated

System Testing is a vague term

- I'm meaning 'system as a whole'
- Stories in combination i.e. functional flows that span stories
- Stories in different orders
- Going beyond the story Acceptance Criteria
- Often 'neglected' in Agile Projects

Tracks v2.3.0 Release Notes

<https://www.getontracks.org/news/comments/release-2.3.0/>

- Changes
- Bug Fixes

Could review code committed since previous release.

Tracks 2.3.0 Released

has been released! This version brings a few small changes to the UI, many bug fixes, and major under-the-hood upgrades and refactors.

Running on Rails 4.1. Ruby 1.8.7 is no longer supported, but this release of Tracks works on Ruby 1.9.3, 2.0.0, and 2.1.

Ability to group todos on the home page by context or by project (using the view menu) also works for tag page, the project page, the tickler and the context page
Ability to change the state of a context to closed
The context menu has been renamed from cz to cs to follow ISO standards
Added a new locale (@AlexStein)
The notes and toggle-collapsed-containers have been moved into the view menu.
Bug fixes

Get Tracks

[Tracks 2.4.1 released](#)

[Report a bug](#)

Github Repository

[tracks](#)

Tracks is a GTD™ v2 application written with Ruby on Rails

[tracksapp.github.io](#)

The official Tracks application on GitHub Pages

[.github](#)

Default community

[@TracksApp](#) on GitHub

Community

Is it a real issue?

I cannot create an Action on a closed context. And I can not change state of a Context to closed when it has open Actions. But I can amend an Action to assign it to a closed context.

- Don't know**
- On a real project I would not have pursued it so far, without finding that out first.**
- It is the type of issue that would be found via**

If it is an issue, when would it be found?

- Structured/Traditional testing**
 - hopefully during a Q&A, clarification when writing 'cases/conditions' from the requirements**
 - but it might take days/weeks to get to that point depending on process e.g.**
 - analyse all test conditions and ideas against requirements and remove ambiguity - might**

If it is an issue, when would it be found?

- Agile
 - hopefully during a story understanding or development session
 - if not then, would it be found from following acceptance criteria?
 - probably not, it would be found in the exploratory testing 'around' the story

Note: questioning and clarifying during analysis is an

Projects make decisions about defects

In this case it seems like a mismatch in states. The constraint is too easy to bypass if it is an important constraint e.g.

Instead of creating the Action on a closed context. I create it on an Active context, then amend the action to be on a closed context.

But, the impact in terms of system reliability,

What Drives Tooling?

- Observation
- Interrogation
- Manipulation
- Admin
 - process support, Evidence Gathering, Note Taking, etc.

Observation

- seeing in real time
- possibility to quick catch issues
- supports monitoring e.g. alerts on errors in logs

Interrogation

- deep dive 'after'
- could be 'seconds' after, but still 'after'
- more data than can be absorbed quickly

Manipulation

- ability to change the system -defaults, static data
- repeat/amend requests
- use the API

Evidence Gathering

- recorded observations
- note taking

What Drives Tooling ... Needs of Testing

For Web

I can test 'blind' and trust the GUI or I can add tooling

- Dev Tools**
- Proxy Tools**

Both allow observation and interrogation. Proxy tool offers scope for easier longer term interrogation and manipulation for replay.

For Technology

Find tools that allow:

- Observation**
- Interrogation**
- Manipulation**
- Support Scripting, extension, apis**
- Record Evidence**

What Did I Not Do/Show? - Admin

- Admin Sessions
- Raising Defects
 - Use existing logs and evidence where possible.
 - Use logs to help recreate
- Clarifying Questions
- Communication
- Uploading Logs to Time Tracker

Bugs

Testing is not all about bugs.

- The most visible output, are bugs.**
- Many projects only care about bugs.**
- Testing is a process of deliberate exploration and coverage.**
- Finding bugs is a side-effect of this process.**

What do I test?

- Requirements, Stories
 - What a User wants to achieve
 - Why they want it
- How it has been implemented
 - Functional Description
 - Technical Risk
- Decisions made during development
- Priorities and Agreed Risk Areas

Taking a System and Beyond View

- assume that testing has been done on the stories and all stories complete in the sprint and release.
- take a System view and ignore specific 'sprint', 'story' or 'release' requirements. View functionality as a whole.
- will not review automated execution. Risk of duplicated effort and scope.

**Q: How does a manager
add value on an Agile
Project?**

A: By looking at the project holistically.

- Teams are focussed on the functionality that they are working on.**
- Teams often don't have time to look beyond.**
- Teams are often not encouraged to think beyond.**

Process risk, can be used to drive testing. Look where others don't.